# Type MS · Pixels, resolution and colour depth

## Answer 1 of 4 · Question 3 part b.ii · 2 marks

2023 JUN · Paper 01 · extract 1 of 5 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3(a) | Award up to **two** marks for a linked explanation, such as:<br>• If the data is incomplete/inaccurate (1) the answers from the model might not be right (1)<br>• If the assumptions the model is based on are inaccurate (1) the answers from the model might be incomplete/inaccurate (1) | Award one mark for identifying a problem and one mark for a reason<br><br>There must be a clear cause and consequence/effect | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(b)(i) | Award **one** mark for any of:<br>• Pixels per inch/ppi (1)<br>• Pixels per centimetre/ppc (1)<br>• Dots per inch/dpi (1) | Ignore units $cm^2$ $inches^2$ | 1 |

# Type MS · Pixels, resolution and colour depth

## Answer 1 of 4 · Question 3 part b.ii · 2 marks

2023 JUN · Paper 01 · extract 2 of 5 · find part b.ii in this mark scheme extract

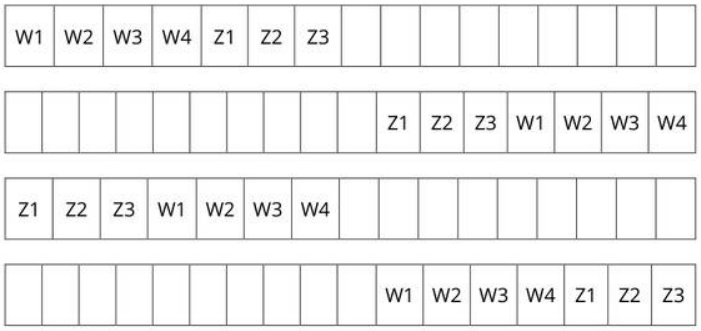
| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(b)(ii) | Award **two** marks for a linked description such as:<br>• Each pixel/colour in the image (1) will require more bits to store (1) | Giving for formula file_size = width*height*colour_depth on its own is not enough | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(c) | Award **two** marks for a linked description such as:<br>• System software is used to maintain/run the operation of the system/hardware (1) whereas application software allows the user to carry out a task (1) | For both marks both categories must be addressed<br><br>Allow clear examples | 2 |

## Type MS · Pixels, resolution and colour depth

### Answer 1 of 4 · Question 3 part b.ii · 2 marks

2023 JUN · Paper 01 · extract 3 of 5 · find part b.ii in this mark scheme extract

<table>
<tr><th>Question number</th><th>Answer</th><th>Additional guidance</th><th>Mark</th></tr>
<tr><td>3(d)</td><td>Award one mark for each of:<br><br>• All file blocks of each file are in numeric sequence/one after another in numeric order (1)<br><br>• All blank space is together at the front/at the end (1)<br><br>Examples:<br><br> 
</td><td>Note: Order of letter and number does not matter. W1 or 1W are equally acceptable</td><td>2</td></tr>
</table>

# Type MS · Pixels, resolution and colour depth

## Answer 1 of 4 · Question 3 part b.ii · 2 marks

2023 JUN · Paper 01 · extract 4 of 5 · find part b.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(e) | Award up to **two** marks for a linked explanation such as:<br>• The file size will not be decreased/will increase (1) because there are very few/no repeating patterns / because a run length of one would be added to each digit (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(f) | Award **one** mark for each correct cell: | | 4 |

| Task | Translator |
|---|---|
| A guessing game that can be used on different computing platforms | Interpreter (1) |
| A screen driver for a new smartphone | Assembler (1) |
| A new version of a spreadsheet program for sale next year | Compiler (1) |
| Control software for an embedded system inside a new washing machine | Assembler (1) |

## Type MS · Pixels, resolution and colour depth

### Answer 1 of 4 · Question 3 part b.ii · 2 marks

2023 JUN · Paper 01 · extract 5 of 5 · find part b.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Pixels, resolution and colour depth

Answer 2 of 4 · Question 1 part g.i · 1 mark

2021 JUN · Paper 01 · extract 1 of 2 · find part g.i in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 1(a) | **The only correct answer is B**<br><br>***A*** *is not correct because computers only understand binary*<br><br>***C*** *is not correct because Hex values will be held as binary so take up the same space*<br><br>***D*** *is not correct because Hex values are held as binary and so take the same time to execute* | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b) | Award **one** mark for each of:<br>• 5<br>• A | Must be in the correct order | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(c) | **The only correct answer is B**<br><br>***A*** *is not correct because this is the number of binary patterns that can be represented by 7 bits*<br><br>***C*** *is not correct because this is the number of binary patterns that can be represented by 9 bits*<br><br>***D*** *is not correct because this is the number of binary patterns that can be represented by 10 bits* | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(d) | Award **one** mark for each nibble<br><br>0110 0101 | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e) | Award **one** mark from:<br><br>• 01000 (1)<br>• 1000 (1) | | 1 |

## Type MS · Pixels, resolution and colour depth

### Answer 2 of 4 · Question 1 part g.i · 1 mark

2021 JUN · Paper 01 · extract 2 of 2 · find part g.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(f) | 7 | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(g)(i) | Award **one** mark from:<br>• The smallest block (of colour) in an image (1)<br>• A single dot in a picture (1)<br>• A picture element (1)<br>• A single element of a digital screen (1) | | 1 |
| 1(g)(ii) | Award **one** mark from:<br>• 8 x 9 (1)<br>• 9 x 8 (1) | Accept 72 | 1 |
| 1(g)(iii) | 4 | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Pixels, resolution and colour depth

### Answer 3 of 4 · Question 3 part b.i · 1 mark

2023 JUN · Paper 01 · extract 1 of 5 · find part b.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 3(a) | Award up to **two** marks for a linked explanation, such as:<br>• If the data is incomplete/inaccurate (1) the answers from the model might not be right (1)<br>• If the assumptions the model is based on are inaccurate (1) the answers from the model might be incomplete/inaccurate (1) | Award one mark for identifying a problem and one mark for a reason<br><br>There must be a clear cause and consequence/effect | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
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## Answer 3 of 4 · Question 3 part b.i · 1 mark

2023 JUN · Paper 01 · extract 2 of 5 · find part b.i in this mark scheme extract

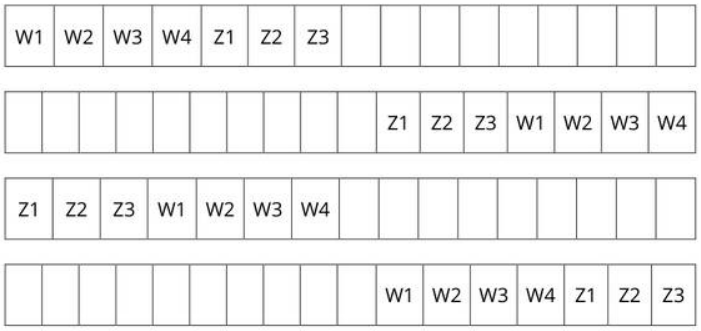
| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(b)(ii) | Award **two** marks for a linked description such as:<br>• Each pixel/colour in the image (1) will require more bits to store (1) | Giving for formula file_size = width*height*colour_depth on its own is not enough | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(c) | Award **two** marks for a linked description such as:<br>• System software is used to maintain/run the operation of the system/hardware (1) whereas application software allows the user to carry out a task (1) | For both marks both categories must be addressed<br><br>Allow clear examples | 2 |

# Type MS · Pixels, resolution and colour depth

## Answer 3 of 4 · Question 3 part b.i · 1 mark

2023 JUN · Paper 01 · extract 3 of 5 · find part b.i in this mark scheme extract

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| 3(d) | Award **one** mark for each of:<br>• All file blocks of each file are in numeric sequence/one after another in numeric order (1)<br>• All blank space is together at the front/at the end (1)<br>Examples:<br>  | Note: Order of letter and number does not matter. W1 or 1W are equally acceptable | 2 |

# Type MS · Pixels, resolution and colour depth

## Answer 3 of 4 · Question 3 part b.i · 1 mark

2023 JUN · Paper 01 · extract 4 of 5 · find part b.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(e) | Award up to **two** marks for a linked explanation such as:<br>• The file size will not be decreased/will increase (1) because there are very few/no repeating patterns / because a run length of one would be added to each digit (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(f) | Award **one** mark for each correct cell: | | 4 |

| Task | Translator |
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## Type MS · Pixels, resolution and colour depth

### Answer 3 of 4 · Question 3 part b.i · 1 mark

2023 JUN · Paper 01 · extract 5 of 5 · find part b.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Pixels, resolution and colour depth

### Answer 4 of 4 · Question 3 part b.i · 1 mark

2025 JUN · Paper 01 · extract 1 of 3 · find part b.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 3(a)(i) | Award **one** mark for each correct hexadecimal character in the correct order:<br><br>• B9 | | 2 |
| Question | Answer | Additional Guidance | Mark |
| 3(a)(ii) | **The only correct answer is B**<br><br>*A is not correct because both are processed at the same speed by a computer*<br>*C is not correct because both use the same amount of storage*<br>*D is not correct because hexadecimal uses number base 16* | | 1 |
| Question | Answer | Additional Guidance | Mark |
| 3(a)(iii) | Award **two** marks for a linked description such as:<br><br>• (logical/arithmetic) right shift (1) by 2 places (1) | | 2 |
| Question | Answer | Additional Guidance | Mark |
| 3(b)(i) | Award **one** mark for:<br><br>Pixel/picture element (1) | Accept additional words.<br>Accept incorrect spelling if word is recognisable as pixel | 1 |

## Type MS · Pixels, resolution and colour depth

### Answer 4 of 4 · Question 3 part b.i · 1 mark

2025 JUN · Paper 01 · extract 2 of 3 · find part b.i in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(b)(ii) | Award **one** mark for each of:<br>• 16x16 pixels (1)<br>• Multiplying (16x16) by 2 bits per pixel in the numerator for colour depth (1)<br>• Dividing by 8 in the denominator to convert to bytes (1)<br>• Dividing 1000 by the number of bytes used for one image (1)<br>Example:<br>$1000 / \frac{16 \times 16 \times 2}{8}$ bytes<br>Allow follow-through | 4 marks for any of:<br>1000 X 8 / 16 x 16 x 2<br>8000 / 256 x 2<br>1000 / 16 x 2 x 2<br>1000 / 16 x 4<br>8000 / 512 | 4 |
| Question | Answer | Additional Guidance | Mark |
| 3(b)(iii) | Award up to **two** marks for a linked explanation such as:<br>• high compression ratio (1) repeated/continuous runs (of the same pixel colour) (1)<br>• requires less storage space (1) encoding 16W for line 1 (1) | Allow valid RLE examples from the image.<br>Explanation can be a combination of different marking points. E.g. MP1a and MP2b | 2 |
| Question | Answer | Additional Guidance | Mark |
| 3(c) | Award **one** mark for each correct cell:<br>Encrypted text / Plain text<br>  RED<br>  COT | Ignore capitalisation<br>Must be correct spelling | 2 |
| | | **Total for question 3** | **14** |



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## Type MS · Pixels, resolution and colour depth

### Answer 4 of 4 · Question 3 part b.i · 1 mark

2025 JUN · Paper 01 · extract 3 of 3 · find part b.i in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|